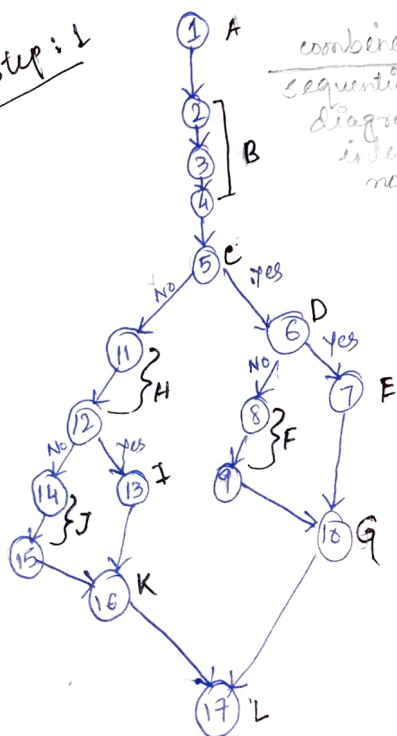


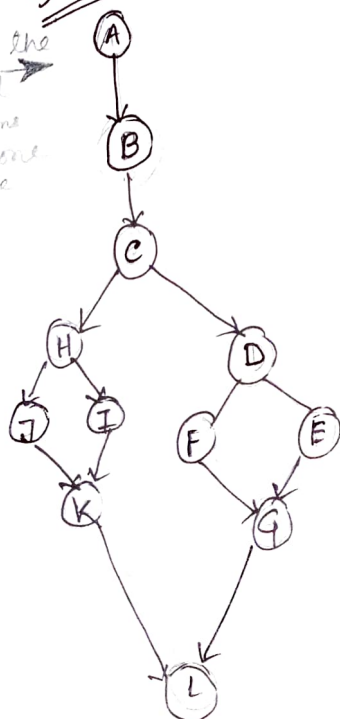
Finding the largest no. using nested if - else

1. function Greatest ()
2. input a
3. input b
4. input c
5. if ($a > b$) {
6. if ($a > c$)
7. print (a is the largest)
8. else
9. print (c is the largest) }
10. end of
11. else {
12. if ($b > c$)
13. print (b is the largest)
14. else
15. print (c is the largest)
16. end if
17. end if

Step 1



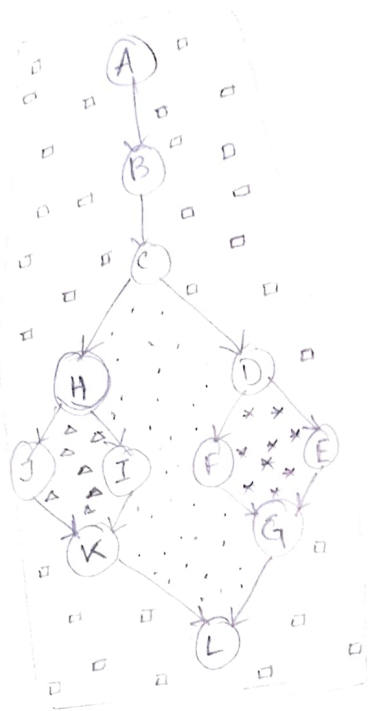
Step 2
combine the sequential diagrams into one node



step 3 Find the cyclomatic Complexity (CC)

method 1:

find closed region. in step 4 : closed region = 4



method 2:

no. of conditions + 1

$$= 3 + 1$$

$$= 4$$

method 3:

$$E - N + 2P$$

$$E = \text{no. of edges} = 14$$

$$N = \text{" " vertices} = 12$$

$$P = \text{no. of ways the program can exit} \\ = \text{only 1 way to exit}$$

$$\therefore CC = E - N + 2P$$

$$= 14 - 12 + 2(1)$$

$$= 2 + 2$$

$$= 4$$

Step 4: Design 4 independent paths.

① A - B - C - H - J - K - L

② A - B - C - H - I - K - L

③ A - B - C - D - F - G - L

④ A - B - C - D - E - G - L

Step 5: Design test cases for each path.